

Introduction

1. What is cloud computing?

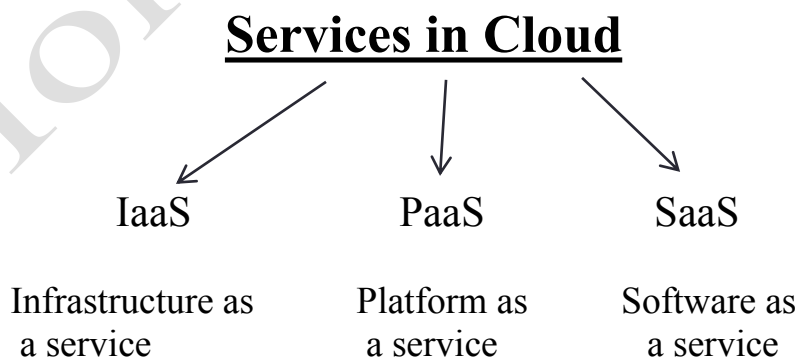
- ◆ Cloud Computing is the On-demand deliver of IT resources over the internet which pay-as-you-go pricing. Instead of buying, owning, and maintaining physical data centers or servers.

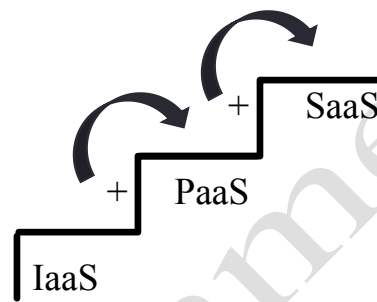
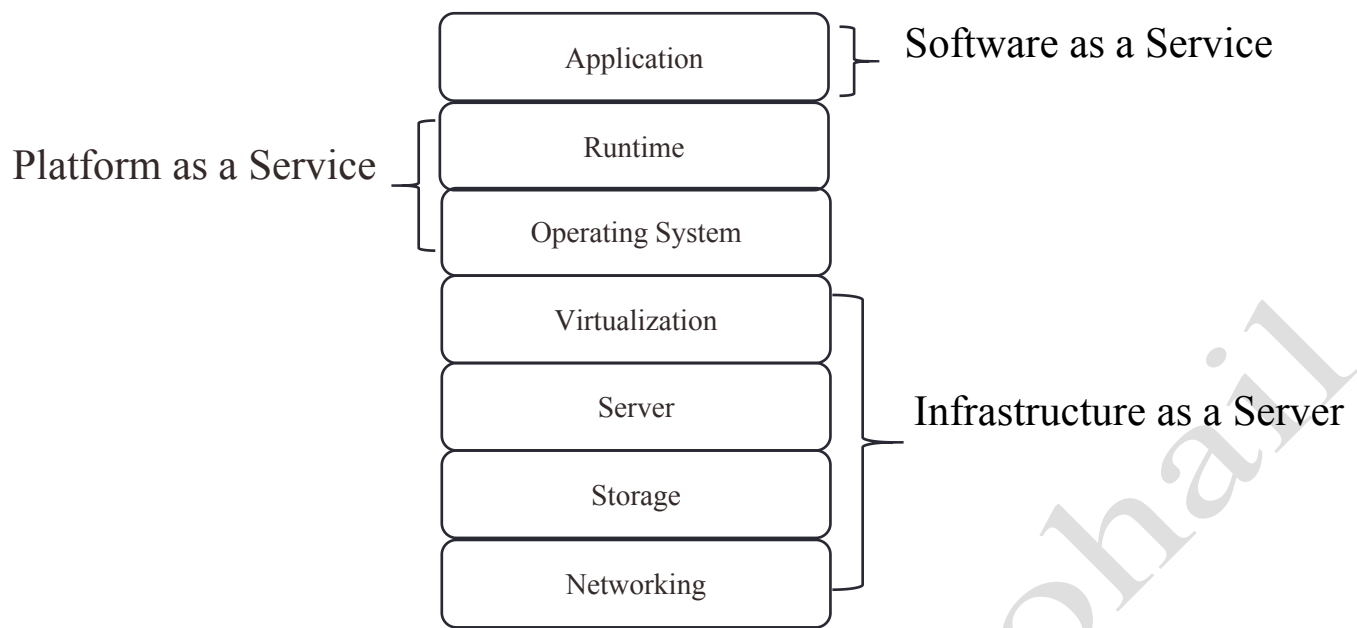
2. On-premise infrastructure VS AWS cloud-base infrastructure

- ◆ In On-premise infrastructure, most of the task/time goes in maintaining the infrastructure, instead of focusing on business needs. Whereas in AWS cloud-based infrastructure, you don't have to worry about maintaining the stability of the infrastructure but you can focus on your business needs

➤ Characteristics of Cloud:

- ◆ On-demand provisioning: Ask for what you need, when you need and don't.
- ◆ Scalability: Scale in/out depending upon the business needs
- ◆ Pay-per-use: Pay for what you have used
- ◆ Measured Service: Every unit of these services is measurable.





1. Public cloud

- ◆ Shared of resources: resources like server and storage are shared among multiple users
- ◆ Cost-effective: pay-as-you-go pricing
- ◆ Scalable: scale in/out depend upon business need

2. Private cloud

- ◆ Dedicated resources: Resources are dedicated to a single organization
- ◆ More control: Organization have more control
- ◆ Cost: Can be more expensive than public cloud

3. Hybrid Cloud?

- ◆ Public Cloud + Private Cloud is called Hybrid Cloud. At least 1 Public and 1 Private.

4. Multi-cloud?

- ◆ If you are using 2 or more Public Cloud is called Multi-cloud.
e.g. Public + Public not Private

1. Introduction AWS

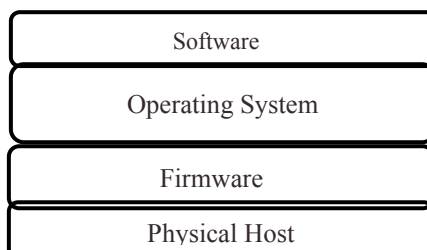
- ◆ Amazon Web Services (AWS) is a bundled remote computing services that provides cloud computing infrastructure over the internet with:
 - ◆ Storage
 - ◆ Bandwidth {Data transfer price}
 - ◆ Customized support for application programming interfaces (APIs) {large Document}

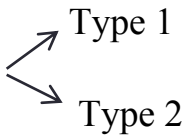
5. Virtualization?

- ◆ It is a technique of splitting as well as combining a physical resource into multiple logical resources. (E.g., Memory, CPU, Disk etc)

6. Hypervisor?

- ◆ It is a Firmware and Software which is used to create and manage virtual machine on a physical host



➤ We have two types of Hypervisor 

- ◆ Type 1 -> Firmware.
- ◆ Type 2 -> Software.

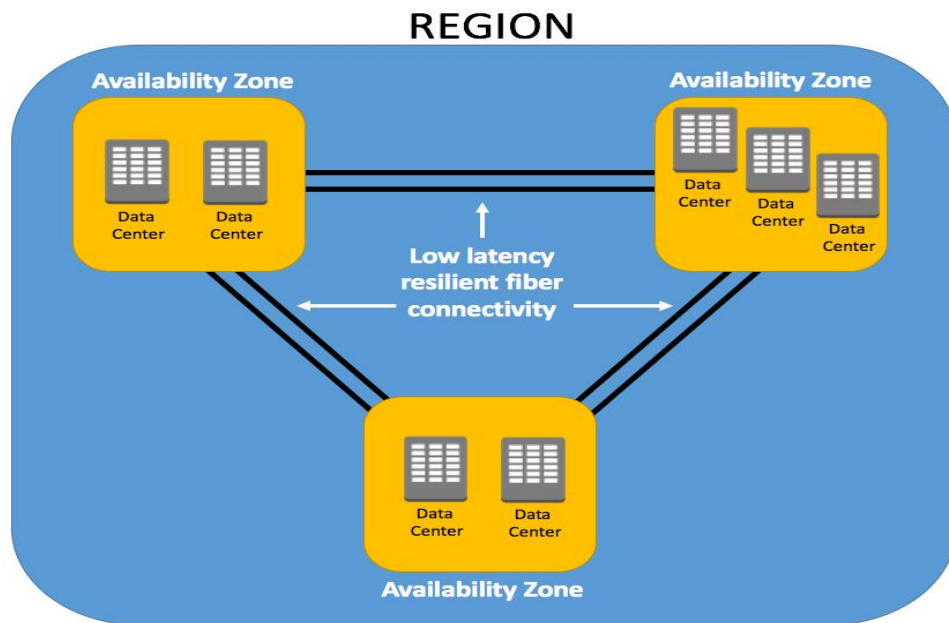
1. Type 1: This type of hypervisor is directly installed on the Bare Metal of Physical Host.
2. Type 2: This type of hypervisor runs on top of an existing operating system already installed on the physical host machine.

➤ **Notes:**

- ◆ AWS is used Custom-designed hypervisor
- ◆ Microsoft Azure is used Hyper-V
- ◆ Google cloud is used KVM (Kernel-base Virtual Machine)
- ◆ VM Ware is used ESXI

AWS Global Infrastructure:

- **Region, Availability Zone and Edge location:** They have 33 region, 105 availability zone and more than 400 Edge location

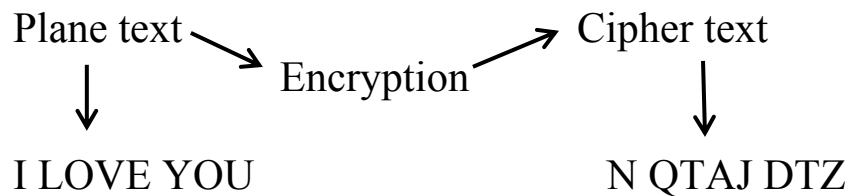


- ◆ **Region:** AWS has a concept of Region, Which is a physical location around the world where we cluster data centers.
- ◆ **Availability Zone:** A collection of data centers within a region. Additionally there are isolated from each other.
- ◆ **Edge Location:** Edge location in AWS store frequently access data as cached data. This is done to improve performance and reduce latency for user. Services like CloudFront use Edge location.

➤ **Shared Responsibility Model:**

1. Encryption

- ◆ Shift each letter by 5 position.

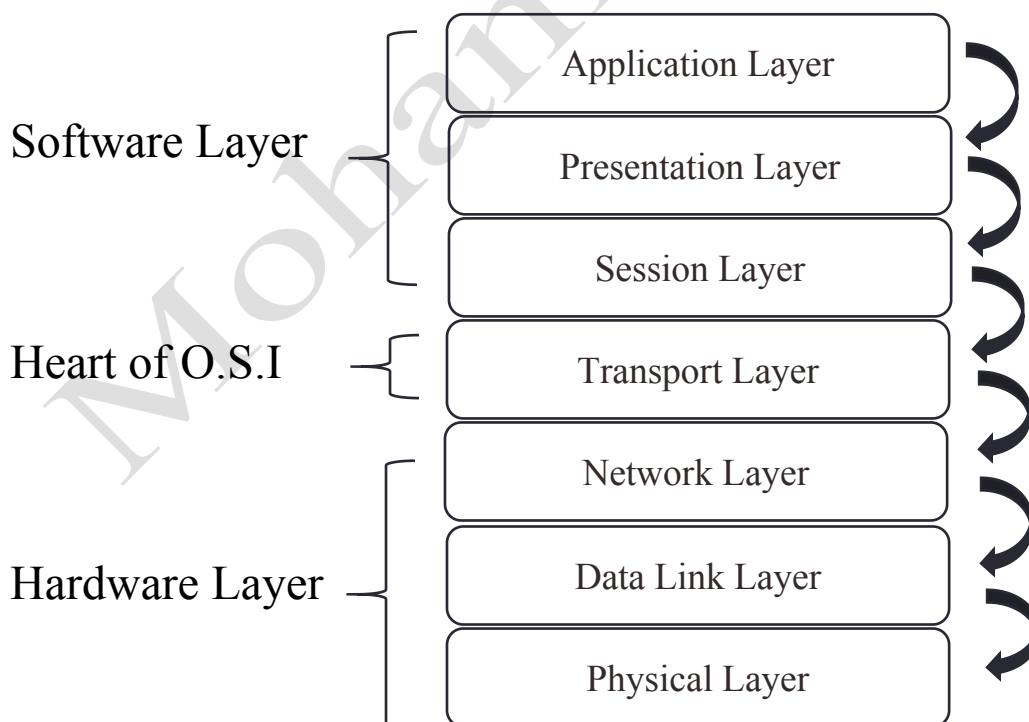


A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

➤ **Differences between GB (gigabyte) and GIB (gibibyte)**

- ◆ 1 GB = 1,000 megabytes (MB)
1,000,000,000 bytes
- ◆ 1GIB = 1024 cubed bytes
1,073,741,824 bytes

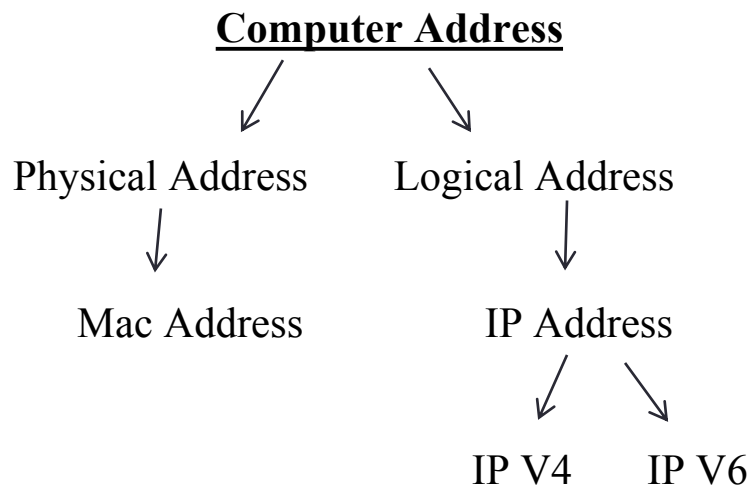
➤ **The Open Systems Interconnection (OSI) model:**



- ◆ Application Layer: Human - Computer interaction layer.
- ◆ Presentation Layer: Ensure the data is in readable format.
- ◆ Session Layer: Maintains connections and it's responsible for controlling port.
- ◆ Transport Layer: Segmentation.
- ◆ Network Layer: Decide Route. Put IP header
- ◆ Data Link Layer: Frame.
- ◆ Physical Layer: Bit cable.

Scope of different services

AZ	Regional	Global
- EC2 Instance - EBS Volume - RDS Instance - Redshift Node - Subnets - Elastic File System	- S3 Bucket - AMI - Snapshot (EBS/RDS) - DynamoDB - VPC - Elastic IP - SQS, SNS - CloudWatch Metrics	- IAM Entities (Users, Groups, Roles & Policies) - Route53 - CloudFront Distribution



➤ **IP Address (IP V4)**

- ◆ 32 Bits Address
- ◆ 4-Octets (Each Octets has 8-bits)
- ◆ Min 0 - 255 Bit Address
- ◆ Binary Address
- ◆ Class A,B,C,D,E
- ◆ Public and Private
- ◆ Network ID + Host ID = IP Address

8-Bit+8-Bit+8-Bit+8-Bit = 32-Bits

192	168	20	40
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4 Octets

➤ 8 Bit Binary Address:

<u>128</u>	<u>64</u>	<u>32</u>	<u>16</u>	<u>8</u>	<u>4</u>	<u>2</u>	<u>1</u>	
1	1	1	1	1	1	1	1	= 255
0	0	0	0	0	0	1	1	= 3
0	0	0	0	1	0	1	0	= 10

➤ Class Types:

Class A - > 0 -- 127

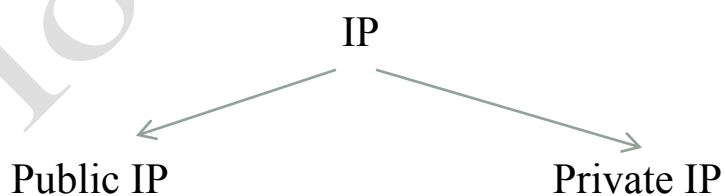
Class B - > 128 -- 191

Class C - > 192 -- 223

Class D - > 224 -- 239

Class E - > 240 -- 255

➤ Public IP and Private IP:



Class A 10.0.0.0 -- 10.255.255.255

Class B 172.16.0.0 -- 172.31.255.255

Class C 192.168.0.0 -- 192.168.255.255

♦ Public IP:

1. Can travel on internet.

♦ Private IP:

1. Can't use WAN.
2. This IP are use in Offices, Library.
3. Broad Cast IP also used Private IP.

➤ Network IP and Host IP:

Class A:	N			
Class B:	N	N		
Class C:	N	N	N	

* N * Stand for Network.
It is fix in Every Class.

Examples:

♦ 172.10.40.60

Network IP : 172.10.0.0
Class : Class B
Broad Cast IP : 172.10.255.255

♦ 205.200.10.20

Network IP : 205.200.10.0
Class : Class C
Broad Cast IP : 205.200.10.255

IAM

Identity and Access Management: It is similar to Active directory on cloud

➤ Why should I use IAM

- ◆ AWS Identity and Access Management is a power full tool for securely managing access to your AWS resources. One of the primary benefits of using IAM is the ability to grant shared access to your AWS account.

➤ Key components of IAM:

- ◆ **Users:** A single entity with user name and password.

Root user: The first user created in AWS account by default it has allow access to all the resources.

IAM user: are created under the root user, by default IAM user has deny access. Max you can created 5000 users. They also have a different URL for an account.

- ◆ **User Groups:** Collections of users who share similar permissions.
- ◆ **Roles:** It is similar to user. It come only with role name and it is attached to AWS services.
- ◆ **Policies:** It is a set of instruction created. Explicity allow or deny Permission. JSON language.

➤ **IAM limit's:**

- ◆ IAM user limits is 5000 per AWS account, you can add upto 10 user at one time.
- ◆ You are also limited to 300 Groups per AWS account.
- ◆ You are limited to 1000 IAM Roles under AWS account.
- ◆ Default limited of managed policies attached to an IAM role and IAM user is 10.
- ◆ IAM user can be a member of 10 Groups (MAX)
- ◆ We can assign two access keys (MAX) to an IAM user.

➤ Amazon Resource Name (ARN):

- ◆ It is a string that uniquely identifies an AWS resource, such as EC2 instance, accounts etc.

➤ Principal:

- ◆ It is a person or application that used an entity {user or role} to send the request.
- ◆ IAM User, Roles, Federated Users and Application are all AWS principal

➤ Authentication:

- ◆ Verifying the identity of a user, often through username and password or multi-factor authentication.

➤ Authorization:

- ◆ Determining what a user is allowed to do after their identity has been verified.

➤ Identity Federation:

- ◆ You can allow users who already have password elsewhere-for example, in your corporate network or with an internet identity provider-to access your AWS account. These users are granted temporary credentials that comply with IAM best practice recommendations. Using identity Federation enhance the security of your AWS account.

1. **Create web site we need important things like:** Text

Images

Videos

2. **Topology**

- ◆ Topology refers to the overall layout and structure of a network. It defines how devices are connections and arranged.

3. (DNS) Domain Name System

- ◆ Human-readable address that translates to a website's numerical IP address.
- ◆ 2 or more parts separated by dots. Example Google.com



4. What is NAT (Network Address Translation), and why is it used?

- ◆ A Network Address Translation (NAT) is the process of mapping an internet protocol (IP) address to another by changing the header of IP packets while in transit via a router. This helps to improve security and decrease the number of IP addresses an organization needs

5. Active Directory

- ◆ It store information about all the users and other resources on a internet
- ◆ List of users, What are they rules and responsibility
- ◆ Active >>> activity of user and Directory >>>> detail of user.

Note:

My account ID is >>>>>>> 102848490242

My account Password is >>>>> Sohail@4928

My account name is >>>>>>>>Mohammed Sohail

IAM user URL <https://102848490242.signin.aws.amazon.com/console>

EC2

➤ **Elastic Compute Cloud (EC2):**

- ◆ Amazon EC2 provides scalable computing capacity in the AWS cloud.
- ◆ We can use Amazon EC2 to launch as many or as few Virtual Services we need. Configure security, Networking and Manage Storage.
- ◆ Amazon EC2 enables us to Scale Up or Scale Down the Instance Capacity.
- ◆ Amazon EC2 has two Storage options i.e., Elastic Block Store (EBS) and Instance Store.
- ◆ Preconfigured Templates are available known as Amazon Machine Images (AMI).

➤ **Difference between EBS and Instance Store?**

- ◆ Instance Store is a laptop storage.
- ◆ Amazon Elastic Block Store (EC2) is directly attached with Host.

EC2 Instance Types:

1. General Purpose: Balance Memory and CPU.
2. Compute Optimized: More CPU than Ram.
3. Memory Optimized: More Ram.
4. Accelerated Computing: Graphics Optimized.

5. Storage Optimized: Low Latency.
6. High Memory Optimized: High Ram, Nitro Hypervisor System.
7. Previous Generation Instance: Previous Instances which we can use.

➤ **General Purpose Instance:**

- ◆ This provide a balance of Compute, Memory and Networking Resources and can be used for a variety of Workloads.
- ◆ 3 Series in General Purpose Instances are as below:

A Series: A1 Instance

B Series: M4 Instance and M5,M5a,M5ab,M5d Instance

T Series: T2,T3,T3a Instance

➤ **Compute Optimized Instance:**

- ◆ Are ideal for compute-bound applications that benefit from high performance processors.
- ◆ It only has C Series which has C4,C5 and C5n

➤ **Memory Optimized Instance:**

- ◆ These are designed to deliver fast performance for workloads that process large data sets in memory.
- ◆ These consists of 3 series and they are R,X,Z.

➤ **Accelerated Computing Instance:**

- ◆ These use hardware accelerators or co-processors to perform some functions such as floating-point number calculations, graphics processing or data pattern, matching more efficiently than is possible in software running on CPU's
- ◆ These consists of 3 Series and they are P,G,F

➤ **Storage Optimized Instance:**

- ◆ These are designed for workloads that require high sequential read write access to very large data sets on local storage.
- ◆ These are optimized to deliver tens of thousands of low latency, random I/O operations per second (IOPS) to application.
- ◆ These consists of 3 Series and they are D,H,I

➤ **High Memory Instance:**

- ◆ This are purpose built to run large-in-memory databases and including production developments of SAP HANA in the cloud
- ◆ These instances are bare metal instances and do not run on a hypervisor.
- ◆ Only available under dedicated host purchasing category (For Min 3 Years term).
- ◆ OS directly on the Hardware.

➤ **Previous Generation Instances:**

- ◆ T1,M1,C1CC2,M2,CR1,CG1,i2,HS1,M3 and R3.
- ◆ These are not deleted instances and we can still purchase these instances

Attribute: It describe individual



EC2 Purchasing Options

Instance Purchasing Options

VM Purchasing Options

All are same

- ◆ There are three ways to pay for EC2 instance i.e. On-demand, Reserved Instance and Spot Instance.
- ◆ Dedicated host and Dedicated instance costs are calculated as per On-demand instance cost and Scheduled instance are built as per reserved instance costs.
- ◆ We can also pay for dedicated host which provides us with EC2 instance capacity on physical server dedicated for our use.



➤ On-Demand Instances:

- ◆ These are virtual servers that run in AWS or AWS Relational Database server (RDS) and are purchased at a fixed rate per hour.
- ◆ AWS recommends using these instances for applications with short term irregular workload that cannot be uninterrupted.
- ◆ They are also suitable for use during testing and development of apps on EC2.
- ◆ With this we can only pay for EC2 Instances we use.
- ◆ Pricing is per instance-hour consumed for each instance, from the time an instance is launched until it is terminated or stopped.
- ◆ Each partial instance consumed will be billed per second for Linux instances and as a full hour for all other instances types.

➤ Dedicated Instances:

- ◆ A Dedicated instances in a vpc is a virtual machine that runs on physical hardware isolated for a single AWS account.
- ◆ Your Server always live here.

➤ Schedule Instances:

- ◆ These enables us to purchase capacity reservation that recur on a daily, weekly or monthly basis with a specified time and duration, for a one-year term
- ◆ We reserve the capacity in advance so that we know it is available when we need it.

➤ Reserved Instances:

- ◆ These provide a significant discount up to 70% compared to on-demand pricing and provide capacity reservation when used in a specific Availability Zone.
- ◆ Reserved Instance give us the option to reserve a database (DB) instances for a one- or three-years term an in turn receive a significant discount compared on demand instances pricing.
- ◆ 3 Types of RI:- Standard, Convertible and Scheduled RI.

Standard RI	Convertible RI	Scheduled RI
• These provide the most significant discount upto 75% off On-Demand and are best suited for steady state usage.	• These provide a discount upto 54% and the capability to change the attribute of RI as long as the exchange results in the creation of reserved instances of greater or equal values.	• These are available to launch within in the time window we reserve....Same as Scheduled Instances.

➤ **Dedicated Host:**

- ◆ An Amazon EC2 dedicated host is a physical server with ec2 instances capacity fully dedicated for our user
- ◆ Dedicated host can help us address compliance requirements and reduce costs by allowing us to use our existing server bound software requirements.
- ◆ Your server lives here and you have control of the physical attribute.

➤ **Spot Instances:**

- ◆ These let us take advantage of unused EC2 capacity in the AWS cloud. These are available up to 90% discount compared to on-demand prices.
- ◆ We can use these for various test and development workloads.
- ◆ We also have option to hibernate, stop or terminate our spot instances when EC2 reclaims the capacity back with two minutes of notice.

AMI

- ◆ Preconfigured Templates are available known as Amazon Machine Images in EC2
 1. Quick Start AMI
 2. AWS Market Place AMI
 3. MY AMI
 4. Community AMI

Two types of Snapshot

1. Volume
2. Instance

Two Types of Security group

1. Security Group: Work at Instance level. Statefull
2. Network Attached Control list (NACL): Its work on Subnet Level. Stateless

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What is a firewall:

A firewall is a network security device or software that monitors and controls incoming and outgoing network traffic based on predefined security rules. It acts as a barrier between a trusted internal network and untrusted external networks.

What is a VPC in cloud:

A Virtual Private Cloud (VPC) is a virtual network within a cloud provider's infrastructure that you define and it allows you to isolate, control and deploy your resources, such as virtual servers and databases.

➤ Port Range numbers:

- ◆ SSH = 22 (Linux Remote Desktop)
- ◆ RDP = 3389 (Windows)
- ◆ HTTP = 80 (For Website, server)
- ◆ HTTPS= 443 (For Website, server with Certificate)

AWS cloud front

Amazon cloud front is a web service that gives business and web application developers an easy and cost effective way to distribute content with low latency and high data transfer speed

Amazon S3 Classes

1. Standard S3

2. Glacier Deep Archive
3. Glacier Flexible Retrieval
4. Glacier Instant Retrieval
5. Standard IA
6. One Zone IA
7. Intelligent Tiring
8. Reduced Redundancy

S3 Versioning: Amazon S3 Versioning to keep multiple versioning of object in one bucket so that you can restore objects that are accidentally deleted.

TCP

Transmission Control Protocall

A-----B

Retransmission Packet

Again an Again if Packet loss

UDP

User Datagram Protocall

NO Retransmission Packet

Real time eetraffic, Live

Implied Router: VPC itself behave aas a implied route

Load balancer : It is a technique used to distribute incomming traffic across multipule server in a network

DevOps:

focuses on making sure that the Development and Operations teams are working together

GIT

Git is a distributed version control system for source code management. Within DevOps, it's used for version control, collaboration, and to automate code deployments via CI/CD pipelines.

What is a branch in Repository:

A branch in a Git repository is a separate line of development that allows multiple developers to work on different features or bug fixes simultaneously without affecting the main codebase

Ansible

What is a playbook in Ansible:

A playbook in Ansible is a YAML file that defines a set of tasks and configurations to be executed on remote servers. It is used for automation and configuration management.

Q, What are some problems that you have faced while working on a project:

Q, What is the difference between Virtualization and containerization:

Q, What problem does Docker solve:

How do you manage multiple containers:

I use container orchestration tools like Kubernetes to manage multiple containers, ensuring high availability, scalability, and load balancing.

What is CI and CD in CICD:

CI (Continuous Integration) is the practice of automatically building and testing code changes frequently. CD (Continuous Deployment/Delivery) is the automated process of deploying code to production after successful CI.

How will you create a CICD pipeline to update the website or app on every commit to a particular branch:

To create a CICD pipeline for this scenario, I would configure the following steps:

1. Set up a version control system (e.g., Git).
2. Use a CICD tool (e.g., Jenkins) to monitor the repository for commits to a specific branch.
3. Configure automated tests to ensure code quality.
4. Build and package the application
5. Deploy the application to a staging environment for further testing (if required).
- + :: 6. If tests pass, automatically deploy the application to the production environment.

Additionally, implement rollback mechanisms and monitoring to ensure the production environment remains stable.

Explain staging, production and testing environment

- Development/Testing Environment: Where developers write and test code.
- + :: • Staging Environment: Pre-production environment for final testing before the live environment.
- Production Environment: Live environment for end-users.

Explain Canary Deployment:

+ :: Canary Deployment is a strategy where a new version of an application is deployed to a small subset of users or servers, allowing for testing and validation. If it performs well, it's gradually rolled out to the entire user base.

What is the biggest issue you faced:

+ :: The most significant challenge I faced was managing a sudden influx of traffic during a product launch. We had to quickly scale our application to handle the load, which involved optimizing database queries and adding more servers.

How do you rollback if something fails:

I roll back by deploying the previous version of the application or by using container orchestration tools to revert to the last stable state. I also ensure comprehensive monitoring and alerts to detect failures early.

How do you automate Deployment:

I automate deployment using scripts and CI/CD pipelines with tools like Jenkins, Travis CI, or GitLab CI/CD. This includes building, testing, and deploying code automatically on every commit. Also continuous monitoring to make sure systems are up and properly working

CHEF

whg

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